

Network Traffic Analysis & Threat Detection

Using Wireshark

Defensive Cybersecurity Project — Blue Team / Network Defense

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1. Project Overview

This project involved performing network traffic analysis in a controlled lab environment to detect suspicious activity, potential attacks, and abnormal communication patterns. Traffic was captured using Wireshark on a Kali Linux monitoring machine, while an Ubuntu virtual machine acted as the target host running an FTP service (vsftpd).

The goal was to simulate the type of packet-level investigation performed by Security Operations Center (SOC) analysts: identifying credential exposure, suspicious DNS activity, and network reconnaissance (scanning) behavior.

2. Environment Setup

Monitoring machine: Kali Linux (Wireshark, running on the "any" interface)

Target machine: Ubuntu (vsftpd 3.0.5 FTP server, account: testuser)

Network: VirtualBox host-only adapter, 192.168.56.0/24

Kali host-only IP: 192.168.56.102 / 192.168.56.103

Ubuntu host-only IP: 192.168.56.101

Kali also retained a separate NAT adapter (10.0.2.15) for internet access, used for DNS resolution during the seeded capture. The two machines were connected on the same host-only network, verified via ping before each capture.

3. Tools & Technologies

Primary tool

- Wireshark — packet capture and protocol analysis

Supporting tools

- Nmap — port scanning (traffic generation)
- nslookup — DNS query testing
- FTP client — credential exchange testing

4. Traffic Captured

Two separate packet captures were produced:

File	Size	Packets	Duration	Description
clean_traffic.pcap	58 MB	70,348	00:07:20	Normal background/browsing traffic (baseline)
seeded_traffic.pcap	385 KB	4,084	00:04:39	FTP login + DNS query + Nmap scan (seeded traffic)

Despite containing the suspicious activity, the seeded capture is far smaller than the baseline — the average packet size was 802 bytes in clean traffic versus just 61 bytes in seeded traffic. This is expected: FTP, DNS, and Nmap exchanges are small, rapid, text-based packets, while ordinary browsing (particularly video) transfers far more data. This size/packet-count contrast is itself a useful observation — small, rapid, repetitive packets are a distinguishing signature of scanning and reconnaissance activity.

5. Protocol Analysis Results

seeded_traffic.pcap was filtered by protocol (http, dns, tcp, ftp) to identify what kinds of communication took place during the seeded activity.

Protocol	Packets found	What it showed
FTP	18 packets	Full login exchange with 192.168.56.101 — USER/PASS commands, directory listing, and session close, all in plaintext.
DNS	6 packets	A query for randomdomain123.xyz (no valid answer returned), plus routine reverse-lookup traffic.
TCP	2,000+ packets	A large burst of SYN packets to 192.168.56.101 across many destination ports, consistent with an Nmap SYN scan.
HTTP	0 packets	No HTTP traffic present — this capture contained no web browsing activity.

The absence of HTTP traffic confirms the capture is isolated to the seeded activity (FTP, DNS, and the port scan) rather than mixed with ordinary browsing, which keeps this file focused for analysis.

6. Suspicious Activity Identified

Finding 1 — Plaintext FTP Credentials

What you saw: While filtering seeded_traffic.pcap by ftp, the FTP login exchange was visible in cleartext. The USER packet showed the username testuser, and the following PASS packet showed the actual password (ubuntu), both fully readable in Wireshark's packet detail pane with no encryption applied.

Why it matters: FTP transmits credentials unencrypted. Anyone capturing traffic on the same network segment can read these credentials directly, without needing to crack or decrypt anything. In a real environment, this could let an attacker steal login credentials just by passively monitoring traffic.

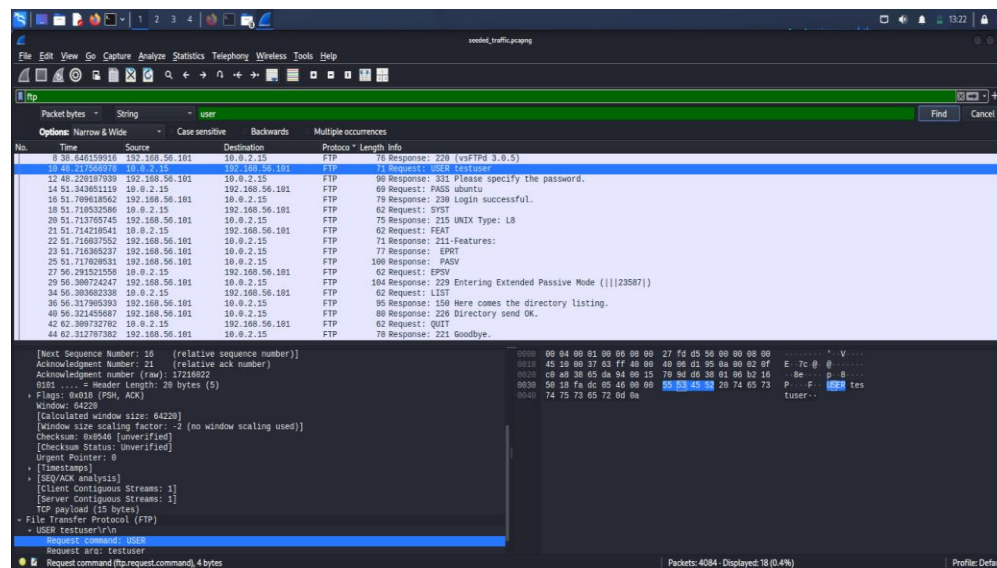


Figure 1a — FTP USER packet showing username "testuser" in plaintext.

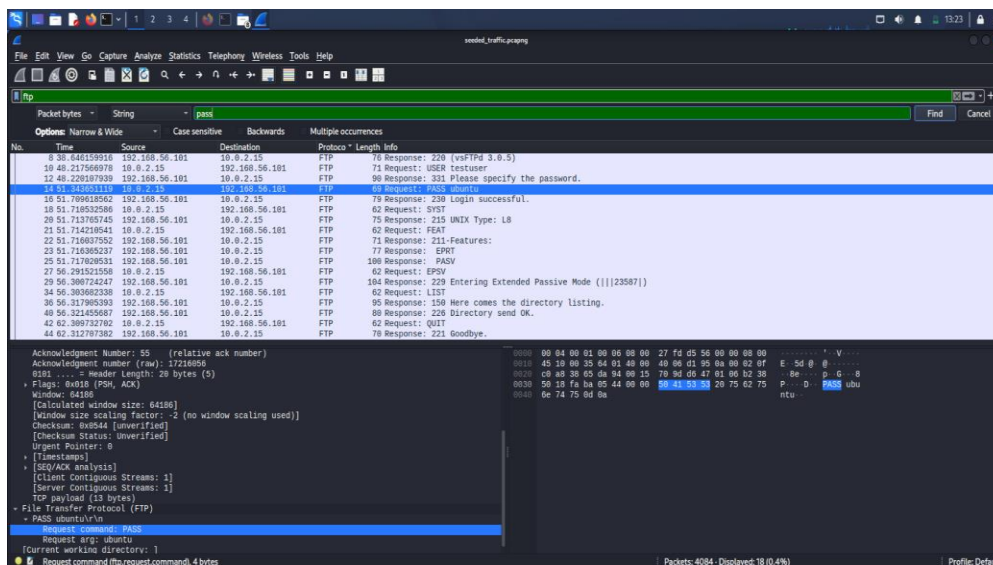


Figure 1b — FTP PASS packet showing password "ubuntu" in plaintext.

Finding 2 — Unrecognized DNS Query

What you saw: Filtering by dns revealed a query for randomdomain123.xyz — a domain with no recognizable business or personal purpose, and one that does not resolve to any real website.

Why it matters: Unexplained DNS lookups to unfamiliar or randomly-generated domains are a known pattern used by malware to reach command-and-control (C2) servers. A single query alone is not proof of compromise, but it is exactly the kind of activity a SOC analyst would flag and investigate further.

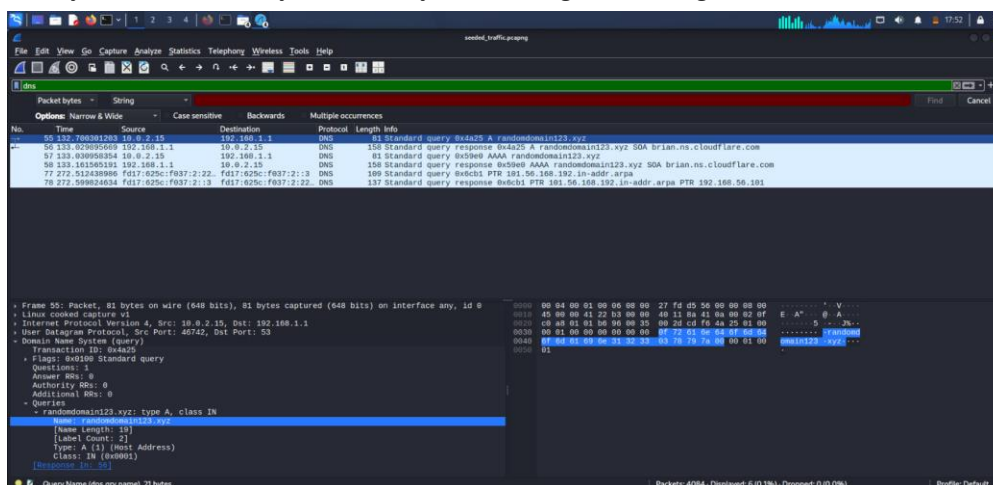


Figure 2 — DNS query to randomdomain123.xyz, expanded in the Queries section.

Finding 3 — Port Scan Detected

What you saw: Filtering by tcp.flags.syn==1 and tcp.flags.ack==0 revealed a burst of over 2,000 SYN packets from Kali to 192.168.56.101, hitting many different destination ports (80, 443, 22, 8080, 5900, 8888, and others) within about one second — the signature of an nmap -sS scan.

Why it matters: This pattern matches network reconnaissance: an attacker systematically probing a target to discover open ports and services before attempting exploitation. A burst of SYN packets to multiple ports from a single source, with no completed connections, is a strong and well-known indicator of scanning activity.

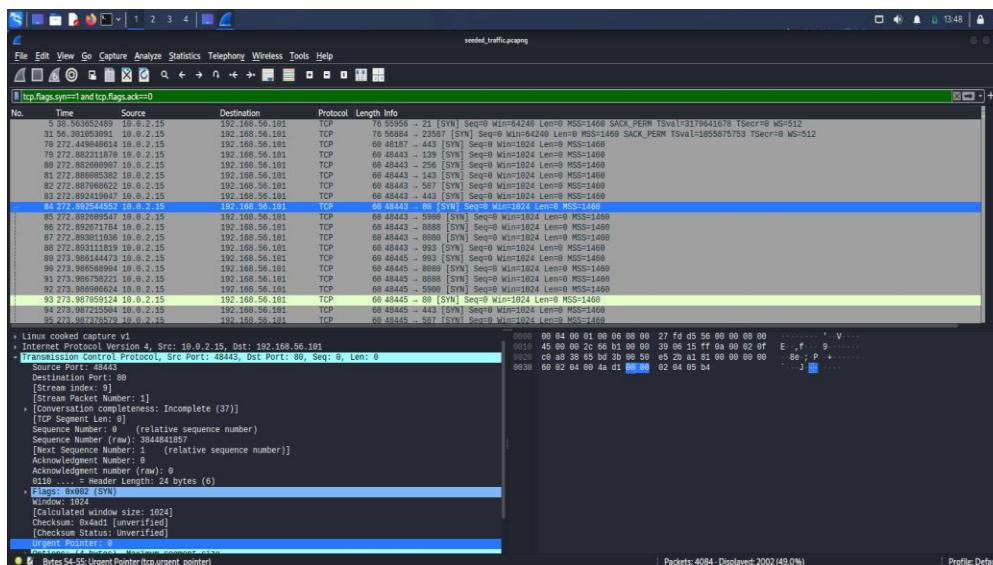


Figure 3 — Burst of SYN packets across multiple destination ports, consistent with an Nmap SYN scan.

7. IP / Domain Investigation

192.168.56.101

This is a private, internal lab IP address (RFC 1918 range). Whois confirms this directly: the lookup returns no organization-specific record, only the IANA reservation entry for the entire 192.168.0.0/16 block, noting these addresses are used by millions of independently operated private networks worldwide and require no coordination with IANA.

VirusTotal, however, does return a result: it reports "10+ detected files communicating with this IP." This is not evidence of anything malicious inside this lab — VirusTotal's dataset is global, built from files uploaded by users worldwide, some of whom may run software (legitimate or otherwise) on their own private networks that happens to use this same common private IP address. Because 192.168.56.101 is private and reused across countless unrelated networks, this flag cannot be attributed to this lab's traffic specifically. No security vendor listed the IP itself as malicious.

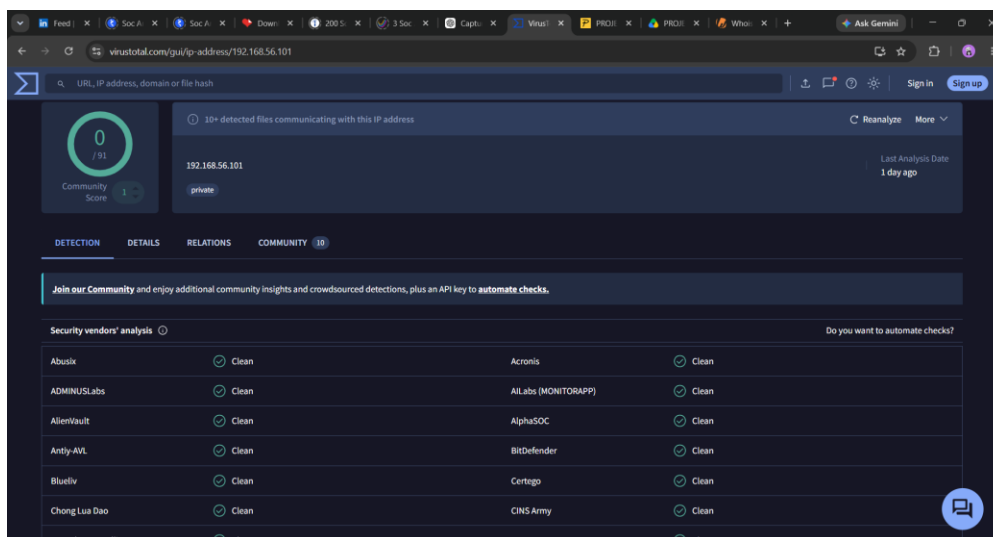


Figure 4 — VirusTotal result for 192.168.56.101, showing 0/91 detections and a note on files communicating with this address.

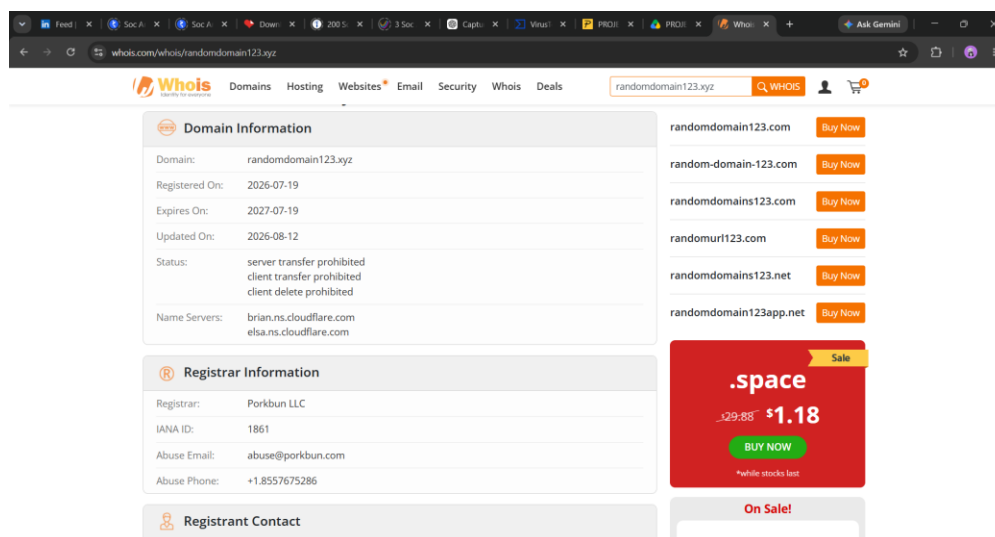


Figure 5 — Whois result for 192.168.56.101, confirming it falls under the IANA-reserved private address block (192.168.0.0/16).

In a real production deployment with a public-facing IP, the same investigation would additionally involve:

- ASN lookup — identifying which organization or ISP owns the IP block
- Geolocation — checking where the IP is physically hosted, to spot unexpected regions
- Known-bad-IP / blocklist checks — cross-referencing threat intelligence feeds (e.g. AbuseIPDB) for prior malicious activity

randomdomain123.xyz

Unlike the initial assumption that this domain would be unregistered, both Whois and VirusTotal confirm it is a real, currently active registration: registered through Porkbun LLC on 2026-07-19, using Cloudflare nameservers, with status flags including transfer/delete protection. VirusTotal additionally tags it "Newly Registered" (24 days old at time of lookup) and reports 0/91 security vendors flagging it as malicious.

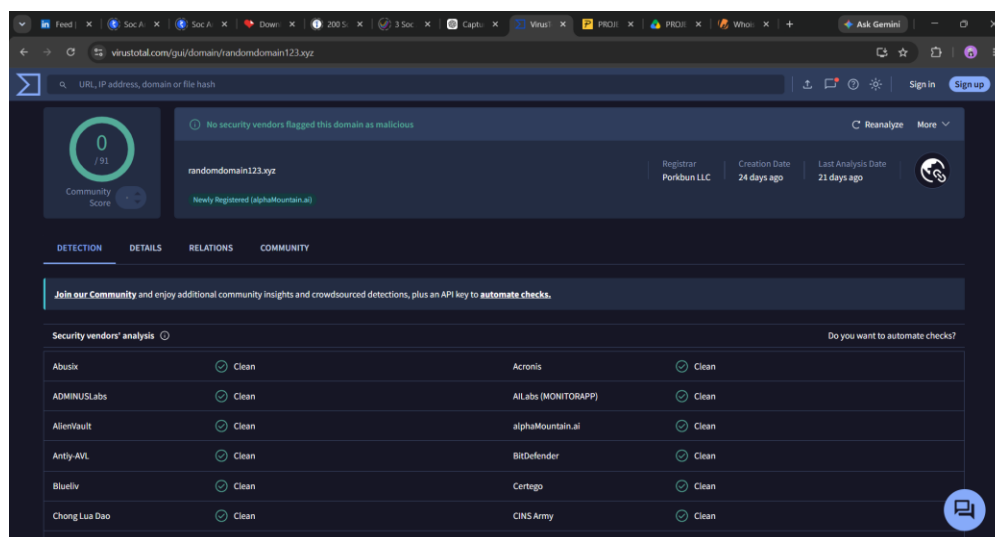


Figure 6 — VirusTotal result for randomdomain123.xyz: 0/91 detections, flagged as "Newly Registered."



Figure 7 — Whois result for randomdomain123.xyz, confirming registration via Porkbun LLC with Cloudflare nameservers.

This is a useful, honest finding in its own right: a domain being registered and clean does not mean it is safe to disregard. In real SOC work, a newly-registered domain (regardless of clean detection results) combined with a randomly-generated, non-branded name is still treated as a mild indicator worth monitoring — attackers frequently register throwaway domains just before use, which means detection engines often have not caught up yet. "0 detections" reflects the current state of security vendor databases, not a guarantee of safety.

8. Risk Assessment

Finding	Risk Level
Plaintext FTP credentials	High
Port scan (reconnaissance)	Medium
Unrecognized DNS query	Low–Medium

Of the three findings, the FTP credential exposure represents the highest risk, as it would allow immediate credential theft by anyone monitoring the network with no additional effort required. The port scan represents a medium risk, since it indicates active reconnaissance against the target but does not itself constitute a breach. The unrecognized DNS query represents a low-to-medium risk on its own, as a single unexplained lookup is inconclusive, but would warrant further monitoring in a production environment given its association with common malware behavior.

9. Security Recommendations

- Replace FTP with an encrypted alternative such as SFTP or FTPS to prevent credential exposure.
- Monitor DNS traffic for queries to newly-registered, unrecognized, or algorithmically-generated domains.
- Deploy network intrusion detection (e.g. Snort, Suricata) to automatically flag SYN scan patterns.
- Restrict and log access to internal services (e.g. FTP) to only trusted hosts, and disable unencrypted services where not required.
- Establish a baseline of normal traffic (as captured in clean_traffic.pcap) to make anomalies easier to spot in future monitoring.

10. List of Suspicious IP Addresses / Domains

- 192.168.56.101 — internal lab target IP. Flagged because it was the destination of the FTP login and the Nmap port scan. Whois confirms it is a private/reserved address; VirusTotal shows no malicious detections, though it lists 10+ unrelated files globally associated with this common private IP.
- randomdomain123.xyz — domain queried via DNS with no clear legitimate purpose in this context. Confirmed via Whois/VirusTotal to be a real, newly registered domain (Porkbun LLC, Cloudflare DNS, 24 days old) with 0/91 malicious detections. Flagged as a suspicious-pattern indicator due to its random-looking name and recent registration, not due to any confirmed detection.

11. Conclusion

This exercise demonstrated core SOC analyst skills: capturing network traffic, filtering by protocol, identifying credential exposure, recognizing suspicious DNS patterns, and detecting reconnaissance activity through SYN scan signatures. The findings reinforce why encrypted protocols, DNS monitoring, and intrusion detection are essential layers of network defense in real production environments.